

Linnéuniversitetet
Kalmar Växjö

Project with Embedded System

Project proposal

Project: Drone

Backup project: Stair-climbing wheelchair



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Contents

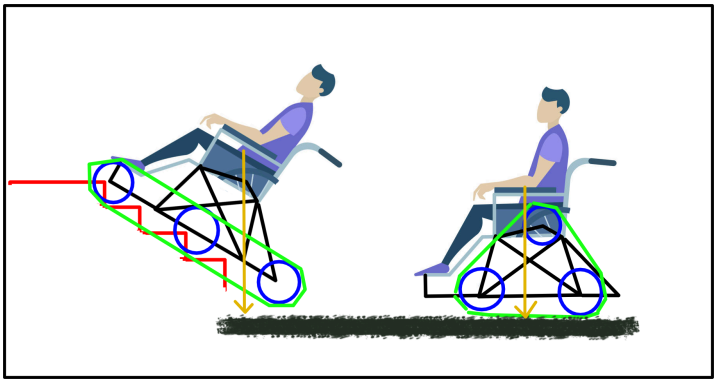
1 Project Idea..... 3
2 Background, Motivation and Proposal..... 4
3 Planning – Milestones and deadlines..... 5
4 Hardware and Budget..... 5
5 Grade Ambition..... 6
6 Sources & References..... 6

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1 Project Idea

Drone: To construct a micro-drone, and make it as stable, cheap, and simple as possible. The main goal is to then use this drone with additional sensors in our thesis project for a concrete real-world application. Our current plan for such a future application is checking for tree damages in the forestry industry (European Spruce Bark Beetle[1] or others[2]). Other possible applications include informed modern agriculture[3]. For this course however, we'll have as a goal to construct a drone with the highest possible carrying capacity and stability for these future uses, and hopefully be able to steer it from a nearby device connected via Bluetooth or WiFi *(both have to be tested)*. We'll start with a RPi Pico W, but might upgrade to a RPi Zero 2 W later.

Wheelchair stair climber: *Wheelchairs are notoriously bad at hinders such as climbing stairs, severely limiting the freedom of mobility impaired. We therefore want to construct a small scale prototype for a wheelchair that has the ability to climb up stairs to increase the autonomy of mobility impaired individuals. The concept would be based on a tri-wheeled caterpillar track. By having the track change shape between a long ski-like shape, and a more compact triangular one when not in stairs, the required flexibility could be achieved. Lengthening the support area this way would ensure the wheelchair to not topple over on inclines much steeper than before.*



Concept sketch by Noa Cluer

2 Background, Motivation and Proposal

Drone:

Due to the nature of us not yet having confirmed a collaborative enterprise, the exact application for a finalized drone is still largely up for debate. Consider the below a general summary of the field.

With the surge of drone development these last couple of years, they are being used in a wide variety of fields, and have a very wide variety of shapes and sizes [4]. This same source also has a wide variety of information on the subject with everything from est. construction costs to different applications and classifications.

Both my father (working with the issue), and [5] explain how work is currently being done; using primarily satellite imagery beetle attacks can be detected. Using these identified attack spots (indicating suddenly dying trees) predictions are made for what areas are going to be affected next. But, these predictions arrive far too late to be very useful, meaning that real-time data from inside the forests in questions are needed, there able to detect based on e.g. bark damage if a tree is affected. Forests are often quite inaccessible, and steering a drone could help examine previously inaccessible areas, and even speed up the process.

Application of drones in modern farming is also a developing area, potentially bringing down costs, and up ecological payoffs due to increased versatility and accessibility. While they could both be used for providing aerial view of fields for everything from damage control to tracking, they can also be used for actually deploying seeds [6] *(in this example used for rice farming)*.

Wheelchair:

There are previous attempts at solving this exact problem [7], [8], with some choosing similar design principles to our suggested solution [9]. But still, the issue remains for many in our society as these only stayed in the project phase, both elderly, handicapped and temporarily incapacitated. Hopefully this new take on the problem could prove less costly, and thereby possible to be made more readily accessible to the general public. The rise of the electric bike also gives us a more industry-wide access to high capacity power sources for these devices, which would be useful to avoid pollution, health hazards, etc.

Worth noting; constructing an entire life-size version of this implementation would prove impossible due to budget and knowledge constraints, so this could at most be implemented at a small demo-scale.

3 Planning – Milestones and deadlines

A preliminary schedule for the project.

Drone:

Noa focus on hardware + report, Samuel focus cad + software + calculations

Week 5: Connect to one electrical motor, and set up working repo
Milestone 1: One propeller works

Week 6: Define list of hardware, design drone case/arms in CAD
Milestone 2: Give it a body

Week 7: Connect to all 4 motors, and mount in/on body (prototype assembly)
Milestone 3: Assembled prototype

Week 8: Test carrying capacity, drone case redesigns
Milestone 4: It flies!

Week 9: Test stabilisation/control + knockover tests
Milestone 5: It flies where we want it to!

Week 10: Testing landing + operating time (power draw) + margin
Milestone 6: We know how long it flies, and where to.

Week 11: Draft report, gather additional sources
Milestone 7: Report written

Week 12: Make build list (costs), finish report
Milestone 8: Instruction manual/instructions done
Milestone 9: Report polished, ready for hand-in.

Wheelchair:

Week 5: System specification, Define list of hardware
Week 6-7: Design blueprints/CAD
Week 8: Hardware component Testing
Week 9: Hardware assembly
Week 10: Testing and finetuning
Week 11: Draft report, gather additional sources
Week 12: Make build list (costs), finish report

4 Hardware and Budget

Module	Quantity	Price / each
DC-motor 50k rpm, 3.7V	4	39,00 SEK
RPi Pico W	1	88,00 SEK
Gyro/IMU: MPU6050	1	59,00 SEK
PLA, wires, resistors, diodes, breadboard, matrix board, etc.	undefined	unknown, but accessible at home/in the lab

Module	Quantity	Price / each
DC-motor 50k rpm, 3.7V	6	39,00 SEK
RPi Pico W *(or any MCU)*	1	88,00 SEK
Caterpillar tracks (PLA)	2	(3D-printed)
Wheelchair model	1	(3D-printed)
hydraulic linear actuators	2	45 SEK
PLA, wires, resistors, diodes, breadboard, matrix board, etc.	undefined	Unknown m, but accessible at home/in the lab

5 Grade Ambition

Passing grade, but really want to extend the project to the thesis project afterwards. Could very well go for a higher grade on the thesis course.

6 Sources & References

[1]https://www.skogsstyrelsen.se/bruka-skog/skogsskador/insektsskador/granbarkborre/tecken-pa-angrepp-av-granbarkborre/
[2]https://www.skogsstyrelsen.se/bruka-skog/skogsskador/svampskador/
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